LLL-S-710H
September 16, 1977
SUPERSEDING
Fed. Spec. LLL-S-710G
April 27, 1971

FEDERAL SPECIFICATION

STEPLADDERS, WOOD

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers, industrial, heavy duty, portable, wood stepladders.

1.2 Classficiation.

1.2.1 Sizes. Wood stepladders covered by this specification shall be of the following sizes, as specified (see 6.2).

Sizes - 3,4,5,6,7,8,10,12,14,16,18 and 20 feet (in length).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-C-650 - Crates, Wood, Open and Covered.
PPP-T-97 - Tape: Pressure-Sensitive, Adhesive, Filament Reinforced.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspections by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

FSC 5440

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standard Institute (ANSI) Standard:

A 14.1 - Safety Code for Portable Wood Ladders.

(Applications for copies should be addressed to the American National Standard Institute, Inc., 1430 Broadway, New York, NY 10018.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 General. In addition to the requirments set forth hereinafter, all ladders shall mee the applicable requirements for type I heavy duty ladders specified in the American National Standards Institute Standard - ANSI A 14.1-1975.

3.2 Hardware.

3.2.1 Fasteners. Nuts shall be square or hexagonal. Square and hexagonal nuts shall have no sharp edges.

3.2.2 Spreader with locking device.

3.2.2.1 Design. The spreader shall be of the pinch-proof design. The locking device shall be integral with the spreader. A metal spreader and locking device shall securely hold the front and back sections in the open position, and shall not permit closing without manual release.

3.2.2.2 Thickness and width. The metal spreader with the locking device shall be of such dimension and strength to with stand the loads imposed, to securely hold the front and back sections in open position.

3.2.2.3 Height above ground level. Spreaders and locking devices for ladders shall be secured on each side, to the back leg and side rail, at not greater than 2/3 the height of ladder. For sizes 16 feet and above, ladders may be equipped with an auxiliary hook-on type metal spreader, at the manufacturer's option. Rivets used in securing spreaders to side rails and back legs shall be sufficiently loose to prevent shearing of rivet heads when ladders are opened or closed.

3.2.3 Protective coating. All ferrous hardware shall be zinc coated except that the edges of spreaders may or may not be coated.

3.3 Bucket shelf. When specified (see 6.2), a bucket shelf shall be provided.

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3.4 Safety shoes. Ladders shall be furnished with safety shoes on both side rails and back legs. The safety shoe shall consist of an angle (L) or channel (U) metal frame with a nonslip pad securely attached to the bottom of the metal frame. The metal frame shall be attached to the side rails and legs by means of bolts, rivets, or screws. The bearing surface of the side rails and legs shall rest on the metal frame. The area of the nonslip pad shall be not less than the cross sectional area of the side rail or leg specified in ANSI A 14.1, table 2 minimum dimensions for type I stepladders. The materials used in the manufacture of the safety shoe shall be the same as used by the manufacturer in his commercial product.

3.5 Placement of steps. Unless otherwise specified (see 6.2), steps shall be placed in accordance with a, or b below.

a. Front and back of steps flush with side rails.
b. Steps flush with front of side rails with protrusion (overhang) to the rear and resulting sharp corners removed.

3.6 Strength and stability requirements.

3.6.1 Compression. When tested as specified in 4.3.1 the ladder(s) shall not crack, break or exhibit any visible weakening of the structure.

3.6.2 Side rail bending. When tested as specified in 4.3.2 the ladder(s) shall not crack, break or exhibit any visible weakening of the structure.

3.6.3 Step bending. When tested as specified in 4.3.3 the ladder(s) shall not crack, break or exhibit any visible weakening of the structure.

3.6.4 Stability. When tested as specified in 4.3.4 all side rails are to remain on the floor during the front, side and rear stability tests. Sliding or skidding shall not be considered cause for rejection.

3.7 Workmanship. The ladders shall be rigid and stable when set upright in the open position. There shall be no visible splits in the wood and all splinters shall be removed (a splinter is defined as any sharp, pointed, piece of wood which cannot be brushed off without bodily injury or damage to clothing). Nail heads shall not be driven deeper than 1/16 in. below the wood surface. Metal parts shall be free from visible corrosion and shall have no sharp edges which will cause bodily injury.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured and tested in accordance with referenced specifications and standards.

4.2 Sampling.

4.2.1 Inspection and quality levels. The Inspection Levels and Acceptable Quality Levels (AQL's), set forth herein, are in accordance with MIL-STD-105.

4.2.2 Visual examination. Ladders shall be examined for defects listed in table I in accordance with MIL-STD-105 using inspection level II. The AQL shall be 4.0 percent defective for major defects and 6.5 percent defective for minor defects.

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TABLE I. Classification of defects end item

Defects
Size not as specified (see 1.2.1).
Wood parts not smoothly machined or dressed on all side (see 3.1).
Wood irregularities not with the specified limits of ANSI A14.1-1975 (see 3.1).
Construction requirements not in accordance with ANSI A14.1-1975 (see 3.1).
Dimensions not in accordance with ANSI A14.1-1975.
Nuts not as specified (3.2.1).
Spreader design, dimensions, or height not as specified (see 3.2.2).
Spreader closes without manual release (see 3.2.2.1).
Protective coating on metal parts not as specified (see 3.2.3).
Bucket shelf missing when specified (see 3.3).
Safety shoes not as required (see 3.4).
Steps not properly installed (see 3.5).
Anti-split device missing (see ANSI A14.1-1975).
Marking missing or not as specified (see ANSI A14.1-1975).
Workmanship not as specified (see 3.7).

4.2.3 Sampling for tests. Sampling for tests shall be in accordance with inspection level S-1. Failure of any sample to pass any test shall be cause for rejection.

4.2.4 Inspection of preparation for delivery. An inspection shall be made to determine whether the preservation, packaging, packing, unitization and marking, as applicable comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table II. For examination of interior packaging the sample units shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. A lot shall be the number of shipping units offered for inspection. Sampling shall be in accordance with MIL-STD-105 using inspection level S-2, AQL 4.0 percent defective.

TABLE II. Classification of preparation for delivery defects

Examine	Defects
Markings	Omitted; incorrect; illegible; improper size, location sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as loose or missing tape on shelf or legs or distortion or container.
	Ladders not adequately taped.
Contents	Number per container is more or less than required. Net weight exceeds requirements.

4.3 Tests.

4.3.1 Compression test. With the ladder in the open position on a level floor a 1000 pound load shall be applied to the center of the top cap, or to the top step if no cap is used. The above load shall be distributed over an area 3-1/2 inches wide by the depth of the step. The load shall be applied for a period of 2 minutes ± 15 seconds to determine conformance with 3.6.1.

4.3.2 Side rail bending test. With the ladder in the open position on a level floor a 1000 pound load shall be applied to the center of the middle step or next higher step. The above load shall be distributed over an area 3-1/2 inches wide by the depth of the step. The load shall be applied for a period of 2 minutes ± 15 seconds to determine conformance to 3.6.2.

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4.3.3 Step bending. With the ladder in the open position on a level floor a 1000 pound load shall be applied to the center of the longest unsupported step and also, not simultaneously, to the longest supported step. The load shall be distributed over an area 3-1/2 inches wide by the depth of the step. The load shall be applied for a period of 2 minutes ± 15 seconds to determine conformance to 3.6.3.

4.3.4 Stability tests. The stability tests involve front, side and rear stability. The ladder is set in the open position on a level floor. A uniformly distributed 250 pound load shall be applied to the top cap or top step where a cap is not used. The ladder is then subjected to a horizontal pulling force in the front, side and rear directions as indicated below to determine conformance with 3.6.4.

Front stability - An 80 pound force is applied horizontally to the front of the ladder at the center of the top step or cap.

Side stability - A 25 pound force is applied is applied horizontally at and toward the side of the ladder at the top step.

Rear stability - A 45 pound force is applied horizontally toward the back of the ladder at the center of the top step.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, or Commercial as specified (see 6.2).

5.1.1 Level A. Each ladder shall be folded (collapsed) with the bucket shelf secured to prevent its projection beyond the outside edges of the back legs. The bucket shelf shall be secured in this poistion, to the back legs by taping, tying with strong twine, or soft wire. The back legs shall be secured to the front side rails with filament reinforced tape conforming to PPP-T-97, type II. The ladders shall than be packed in quantities not to exceed the weight limitation, in crates conforming to PPP-C-650, type II, style A. Waterproofing of contents shall be in accordance with the appendix to the crate specification.

5.1.2 Commercial. The ladders shall be preserved and packed low normal commercial practice. The complete pack shall be designed to protect the item against damage during shipment, handling and storage, to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity and shall be acceptable by common carrier under the National Motor Freight Classification or Uniform Freight Classification.

5.2 Unitization. When shipments to Government depots are full car or truck load, the packed ladders shall be unitized for shipment and handling in accordance with normal commercial practice. The unitized load shall not exceed 2,500 lbs. in weight, 63 inches in height, and 45 inches in width.

5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2 as specified (see 6.2).

5.3.1 Civil agencies. Packages, shipping containers and unitized loads (when applicable) shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. Packages shipping containers and unitized loads (when applicable) shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.

6.1.1 Type I. Heavy duty ladders are intended for general use under adverse conditions where safety is of prime importance.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and size required (see 1.2.1).
(c) Whether bucket shelf is required (see 3.3).
(d) Placement of steps, when specified (see 3.5).
(e) Packing level required (see 5.1).
(f) Whether civil or military marking is required (see 5.3).

Custodian:

Air Force - 99
Army - ME

Preparing Activity:

GSA-FSS

User interest: MC

Review interest: CS

MILITARY COORDINATING ACTIVITY:

Navy - YD

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 40 cents each.

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